

Michael Vinh Pham

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Education

Sep. 2025 - Present	New York University , GSAS Non-Degree Student, Mathematics and Economics Coursework: Graduate Convex Optimization, PhD Macroeconomics
Sep. 2021 - May 2025	Swarthmore College , B.A., Mathematics (3.7 GPA) and Economics (3.8 GPA) -Sigma Xi Research Honor Society, 2-year D3 Walk-on XC/Track Student-Athlete, Swarthmore College Computer Society

Research Assistance

June 2025 - Present	Research Analyst , <i>Federal Reserve Bank of New York</i> - DSGE Team (Monetary and Macroeconomic Studies), Advised by Dr. Marco Del Negro -Implemented algorithms to solve Heterogeneous Agent New Keynesian (HANK) models (dynamic programming, stochastic control at the agent level with general-equilibrium iteration over aggregate variables) -Improved Representative Agent New Keynesian (RANK) estimation: derived higher-order and faster Hessian finite difference equations, more accurate convex optimization methods, stabilized numerical operations for ill-conditioned matrices -Ran r-Star Models for New York Fed President, New York Fed's FOMC Black Books -Open source contributor to FRBNY's DSGE package DSGE.jl
May 2024 - July 2024 (Full-time)	Economics Research Assistant for Dr. Stephen O'Connell , <i>Swarthmore Economics</i>
Sep. 2024 - May 2025 (Part-time)	-Implemented Overlapping generations heterogeneous agent (OLG) models (finite-horizon stochastic optimal control problems w/ equilibrium constraints) -Managed research team of two undergraduate researchers
Jan. 2024 - May 2025 (Part-time)	Economics Research Assistant for Dr. Stephanie Kestelman , <i>Harvard Economics</i> -Created ML pipelines (large-scale data scraping, text unsupervised learning, sentiment analysis, LLM fine-tuning) to understand what was being discussed from LA City Planning Commission hearings -Trained models to perform semantic segmentation on Google Streetview imagery to analyze impact of multi-family housing on urban greenery (working paper)
June 2023 - Sep. 2023 (Full-time)	Machine Learning Research Assistant for Dr. Ryan Kastner and Dr. Curt Schurgers , <i>UC San Diego CSE (REU)</i> -Researched and experimented with CV techniques to semantically segment UAV, satellite imagery to track mangrove tree deforestation -Created pipeline to source satellite data, perform GIS processing, feature engineering, and run ML models with GeoPandas, PyTorch, Hugging Face -Improved lab's mangrove satellite identification from 93% to 98% pixel-wise accuracy
Feb. 2023 - June 2023 (Part-time)	Machine Learning Research Assistant for Dr. Xiaodong Qu , <i>Swarthmore CS</i> -Read 100+ ML for EEG classification papers and second-authored systematic review accepted to KDD and HCII conferences
Nov. 2022 - May 2023 (Part-time)	Bioinformatics Research Assistant for Dr. Renuka Nayak , <i>UCSF School of Medicine</i> -Wrote R, Python Tensorflow scripts to analyze high-dimensional, sparse microbiome arthritis development data -Developed new ML architecture achieving significant MCC score info gain over existing models

Teaching Assistance

Fall 2024, Spring 2025	Social Sciences Quantitative Lab (Stata)
Fall 2024	Intermediate Microeconomics
Spring 2024, Fall 2023	Introduction to Econometrics
Spring 2024	Public Economics
Fall 2022	Introduction to Economics

Other Work Experience

May 2022 - Aug. 2022 (Full-time)	Quantitative Intern, Karner Blue Capital
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Personal

Programming Languages	<i>Extensive Use:</i> Python, Julia. <i>Been a While:</i> C++/C, Matlab, Stata, SQL, JavaScript, x86 Assembly
Citizenship	United States
Languages	English (Native), Vietnamese (C1), Spanish (B2)Mandarin (A1)

Selected Works

2025	Human Capital Formation and Inequality in a Heterogeneous Agent Dynastic Model with Stephen O'Connell, <i>Working Paper</i> Should public spending on early-age human capital formation be protected during fiscal consolidation? We build a heterogeneous agent dynastic macroeconomic model in which parental inputs combine with public resources to produce skills that become productive when children reach adulthood. Private inputs face borrowing constraints and potentially parental misperceptions of the human capital production function. We calibrate the model to match these distortions in the data, then quantify how reductions in public support affect lifetime income and wage inequality within affected cohorts, and the intergenerational transmission of inequality.
2025	Barking up the Wrong Tree? Multifamily Housing and Tree Loss: Evidence from Los Angeles, Working Paper This paper evaluates how the construction of multi-family properties in Los Angeles affect the neighborhood characteristic nearby as defined as greenery coverage. I create address-level panel data by employing computer vision techniques applied to street-level imagery and use difference-in-differences estimators. I find that multi-family development has no negative effect on nearby tree and greenery coverage and may even contribute to a modest increase, with estimates indicating a statistically significant 2.3% rise in total greenery (trees and shrubs) coverage.
2023	A Review of Machine Learning Algorithms for EEG Datasets with Nathan Koome Murungi, Xufeng Caesar Dai, and Xiaodong Qu <i>Accepted: 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (2023)</i> In this paper, we present a systematic literature review that explores the utilization of machine learning (ML) algorithms for analyzing datasets from Electroencephalography (EEG) based Brain-Computer Interfaces (BCIs). Our primary aim is to provide computer science students with a comprehensive and accessible overview of the role of machine learning in EEG analysis. By synthesizing and organizing recent research from 2020 onwards, our objective is to empower the target audience to develop a solid foundational understanding of the current state of ML-EEG research. Through this work, we intend to enhance the accessibility and comprehension of ML-EEG studies and contribute to advancing BCI technology.